Table 9.5 Factors affecting Basal Metabolic Rate (BMR)

S.no	Factor	Effect on BMR
1	Body composition	More the lean body mass, higher is the BMR. This is due to greater metabolic activity in these tissues when compared to bones and fat. Men with a high proportion of muscle mass or lean body mass have a higher BMR than women
2	Fever	Fever raises the BMR. There is a 7% increase in BMR for each degree rise in body temperature in Fahrenheit and 13% increase for every degree Celsius rise in body temperature
3	Stress	Raises BMR.
4	Smoking and Caffeine	Increases BMR
5	Hyperthyroidism (Oversecretion of thyroxine)	The basal metabolic rate is elevated as much as 50-70%.
6	Growth	In tall people BMR is higher.
7	Pregnancy	During the last trimester of pregnancy, basal metabolic rate is increased by 15 - 25% as there is a increase in muscle mass of uterus, size of mammary gland, foetal mass and placenta, cardiac work and respiratory rate.
8	Fasting/Starvation	Lowers BMR
9	Hypothyroidism (under secretion of thyroxine)	The basal metabolic rate is decreased by 30%
10	Age	Lean body mass diminishes with age slowing the BMR. BMR is higher in infants and young children than in adults
11	Undernutrition	Prolonged undernutrition lowers the BMR.
12	Climate	In persons living in tropical climates,BMR is about 10% less than those living in temperate zones.
13	Sleep	BMR is reduced by 5%.

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